

APEX UPI – High-Concurrency PSP Switch & Ledger Engine

1. Introduction

The APEX UPI platform is a distributed payment processing system designed to simulate the architecture of a modern Unified Payments Interface (UPI) ecosystem.

The platform acts as a Payment Service Provider (PSP) Switch responsible for:

- Payment processing
- Balance inquiries
- VPA validation
- Ledger maintenance
- Reconciliation
- Recovery workflows
- Audit tracking

The system uses microservices and event-driven communication to achieve scalability and fault tolerance.

2. Purpose

The purpose of this document is to define the software requirements for the APEX UPI platform.

This document serves as the baseline for:

- Design
- Development
- Testing
- Deployment
- Maintenance

3. Scope

The system shall provide:

Transaction Processing

- UPI Payment Initiation
- UPI Payment Completion
- Transaction Tracking

Banking Operations

- Account Debit
- Account Credit
- Balance Verification

Validation

- VPA Validation
- Fraud Validation
- Limit Validation

Ledger Management

- PSP Ledger
- NPCI Ledger

Recovery

- Reversal Processing
- Timeout Recovery

Monitoring

- Dashboard Services
- Health Monitoring

Auditing

- Event Logging
- Transaction History

4. Definitions

Term	Meaning
PSP	Payment Service Provider
NPCI	National Payments Corporation of India
CBS	Core Banking System
VPA	Virtual Payment Address
RRN	Retrieval Reference Number
TPAP	Third Party Application Provider
DLQ	Dead Letter Queue
Ledger	Financial transaction record
Reversal	Transaction rollback

5. System Overview

High-Level Flow

TPAP
↓
TPAP Ingress Service
↓
Rules Validation Service
↓
Transaction Orchestrator
↓
NPCI Adapter
↓
NPCI Router
↓
CBS Service
↓
Ledger Services
↓
Audit Service

6. Stakeholders

Primary:

Customers

Initiate payment transactions.

PSP Operator

Operates payment switch.

Bank

Maintains customer accounts.

NPCI

Routes transactions.

Secondary:

Operations Team

Monitors system.

Audit Team

Reviews logs.

Compliance Team

Reviews financial records.

7. Functional Requirements

FR-1 Payment Initiation

Description

System shall accept payment requests from TPAP applications.

Input

```
{  
  "payerVpa": "alice@sbi",  
  "payeeVpa": "bob@hdfc",  
  "amount": 1000  
}
```

Processing

1. Validate request
2. Generate Transaction ID
3. Generate RRN
4. Publish Kafka Event

Output

```
{  
  "txnId": "TXN12345",  
  "status": "INITIATED"  
}
```

Priority

Critical

FR-2 VPA Validation

Description

System shall validate payer and payee VPAs.

Validation Rules

- VPA must exist
- VPA must not be blacklisted
- VPA must be active

Failure Response

```
{  
  "status":"FAILED",  
  "reason":"INVALID_VPA"  
}
```

FR-3 Fraud Validation

Description

System shall perform fraud screening.

Rules

- Blacklisted User
- Excessive Transactions
- Suspicious Patterns

Outcomes

APPROVED

REJECTED

REVIEW_REQUIRED

FR-4 Balance Verification

Description

System shall verify account balance before debit.

Validation

Balance $\geq$ Transaction Amount

Failure

INSUFFICIENT_FUNDS

FR-5 Debit Processing

Description

System shall debit payer account.

Steps

1. Receive debit request
2. Validate balance
3. Update account
4. Record ledger entry

Status

DEBIT_SUCCESS

DEBIT_FAILED

FR-6 Credit Processing

Description

System shall credit payee account.

Steps

1. Receive credit request
2. Update balance
3. Record ledger entry

Status

CREDIT_SUCCESS

CREDIT_FAILED

FR-7 Transaction Completion

Description

System shall complete transaction after successful debit and credit.

Final Status

SUCCESS

FR-8 Transaction Reversal

Description

System shall reverse transactions when failure occurs after debit.

Trigger Conditions

- Credit Failure
- Timeout
- System Failure

Process

Debit Success

Credit Failure

↓

Reversal

↓

Refund

Final Status

REVERSED

FR-9 Balance Inquiry

Description

System shall provide account balance.

Request

```
{  
  "vpa": "alice@sbi"  
}
```

Response


```
{  
"balance":15000  
}
```

FR-10 Transaction Status Inquiry

Description

System shall provide transaction status.

Statuses

INITIATED
VALIDATED
PENDING
DEBITED
CREDITED
COMPLETED
FAILED
REVERSED

FR-11 Ledger Recording

Description

System shall maintain immutable ledger entries.

Ledger Types

- Debit Ledger
- Credit Ledger
- Reversal Ledger

FR-12 Audit Logging

Description

System shall maintain audit logs.

Logged Data

- Timestamp
- Service Name
- Event Name
- Transaction ID

FR-13 Dashboard Monitoring

Description

System shall expose monitoring APIs.

Dashboard Views

- Executive View
- Live Transactions
- Kafka Monitor
- Ledger View
- Reconciliation View
- Audit View

FR-14 Reconciliation

Description

System shall identify inconsistencies between ledgers.

Checks

- Missing Credit
- Missing Debit
- Duplicate Transaction

Actions

- Alert
- Recovery
- Manual Review

FR-15 Recovery Engine

Description

System shall recover incomplete transactions.

Recovery Cases

- NPCI Timeout
- CBS Timeout
- Kafka Failure

Actions

Retry

Reconcile

Reverse

8. Non-Functional Requirements

NFR-1 Availability

System availability shall be:

$\geq 99.9\%$

NFR-2 Scalability

System shall support horizontal scaling.

Components

- Kafka
- PSP Services
- Ledger Services

NFR-3 Performance

Payment Processing

Target:

< 5 seconds

Balance Inquiry

Target:

< 2 seconds

NFR-4 Reliability

System shall guarantee:

- No data loss
- Event persistence
- Transaction traceability

NFR-5 Fault Tolerance

System shall continue operation despite:

- Service failure
- Kafka failure
- Database restart

NFR-6 Observability

System shall provide:

- Metrics
- Logs
- Health checks
- Transaction timelines

NFR-7 Maintainability

Microservices shall be independently deployable.

NFR-8 Auditability

Every financial transaction shall be auditable.

9. External Interface Requirements

REST APIs

Payment API

POST /payment/initiate

Balance API

POST /balance/inquiry

Status API

GET /status/{txnId}

Kafka Interface

Topics

upi.transactions.initiated

upi.cbs.debit

upi.cbs.credit

upi.cbs.reversal

upi.transactions.status

Database Interface

PostgreSQL

Stores:

- Accounts
- Ledgers
- Audit Records

Cache Interface

Redis

Stores:

- Temporary state
- Session data
- Recovery checkpoints

10. Use Cases

UC-1 Make Payment

Actor

Customer

Flow

1. Enter Payee VPA
2. Enter Amount
3. Submit Payment
4. System Validates
5. Debit Account
6. Credit Payee
7. Display Success

UC-2 Check Balance

Actor

Customer

Flow

1. Request Balance
2. Query CBS
3. Return Balance

UC-3 View Status

Actor

Customer

Flow

1. Enter Transaction ID
2. Retrieve Status
3. Display Result

11. Data Requirements

Account Entity

VPA

Name

Bank

Balance

Transaction Entity

Transaction ID

RRN

Amount

Status

Timestamp

Ledger Entity

Ledger ID

Txn ID

Type

Amount

Status

12. Security Requirements

Authentication

Authorized service access only.

Authorization

Role-based access.

Data Integrity

Transaction tampering prevention.

Audit Compliance

All changes logged.

13. Reliability Requirements

System shall:

- Persist all events
- Maintain transaction consistency
- Recover from failures

14. Availability Requirements

System shall support:

- Multi-service deployment
- Recovery workflows

15. Assumptions

- Kafka available
- PostgreSQL available
- Redis available
- Network connectivity maintained

16. Constraints

- 5 VM deployment
- Simulated NPCI
- Simulated CBS

17. Acceptance Criteria

Requirement	Acceptance Criteria
Payment	Successful completion
Debit	Correct balance deduction
Credit	Correct balance addition
Reversal	Funds restored
Ledger	Entry recorded
Audit	Event captured
Dashboard	Data visible
Recovery	Failed transaction recovered

18. Traceability Matrix (RTM)

Business Objective	Functional Requirement
Payment Processing	FR-1, FR-5, FR-6
Validation	FR-2, FR-3
Balance Inquiry	FR-9
Monitoring	FR-13
Recovery	FR-15
Reconciliation	FR-14
Auditing	FR-12
Ledger Management	FR-11

Conclusion

The APEX UPI SRS defines all functional and non-functional requirements necessary to build a distributed, event-driven PSP Switch capable of processing UPI transactions with ledger consistency, fault tolerance, recovery mechanisms, reconciliation workflows, and operational observability. This document serves as the baseline for the HLD, LLD, development, testing, and deployment phases of the project.